Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.17	20.23	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.02	+0.3%
power (W), mean	628	638	+1.7%
power (W), peak	651	672	+3.2%
energy (Wh)	211	215	+2.0%
energy, parked workers still on at idle power (Wh)	211	215	+2.0%
CPU used (cores), mean	0.860	1.13	+31.5%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.036	0.047	+31.5%
energy per core-hour (Wh)	730	564	-22.7%
node-hours per core-hour	6.98	5.31	-24.0%
pending pods, pod-minutes	0.000	0.000	0
pending pods, peak	0.000	0.000	0
HPA replicas, mean	9.16	9.21	+0.6%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.517	0.517	+0.0%
requests timed	1,070	1,120	+4.7%
time over the response line (% of samples)	0.280	0.000	-100.0%
response time (ms), mean	125	76.05	-39.3%
response time (ms), median	112	68.40	-39.0%
response time (ms), 95th percentile	253	105	-58.4%
response time (ms), 99th percentile	351	122	-65.3%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.5973	6.0238	-20.7%	[-3.4026, +0.2544]	EVEN, no significant difference
kwh_per_core_hour	0.7917	0.6356	-19.7%	[-0.3278, +0.0210]	EVEN, no significant difference
utilisation	0.0355	0.0466	+31.3%	[-0.0004, +0.0235]	EVEN, no significant difference
pending_pod_minutes_per_hour	0.0000	0.0000	n/a	[+0.0000, +0.0000]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	1.4634	1.4942	+2.1%	[-2.2068, +2.9342]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.391 0.42 0.445 0.468 0.487 0.504 0.52 0.533 0.545 0.555 0.56 0.566 0.57 0.575 0.58 0.584 0.586 0.588 0.589
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

```
restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
```

Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	618	617	6.00	6.00	0.000	0.000	0.597	0.575

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	618	621	6.00	6.00	0.000	0.000	0.585	0.668
2	6.00	6.00	617	624	6.00	6.25	0.000	0.000	0.575	0.741
3	6.00	6.00	620	628	6.00	7.00	0.000	0.000	0.619	0.860
4	6.00	6.00	634	644	9.50	9.25	0.000	0.000	1.02	1.30
5	6.00	6.00	632	648	10.00	10.00	0.000	0.000	0.989	1.38
6	6.00	6.00	633	646	10.00	10.00	0.000	0.000	1.01	1.38
7	6.00	6.00	643	664	10.00	10.00	0.000	0.000	1.27	1.83
8	6.00	6.00	645	668	10.00	10.00	0.000	0.000	1.32	1.95
9	6.00	6.00	643	665	10.00	10.00	0.000	0.000	1.26	1.84
10	6.00	6.00	644	660	10.00	10.00	0.000	0.000	1.30	1.73
11	6.00	6.00	617	621	10.00	10.00	0.000	0.000	0.578	0.664
12	6.00	6.00	617	623	10.00	10.00	0.000	0.000	0.604	0.721
13	6.00	6.00	623	628	10.00	10.00	0.000	0.000	0.734	0.850
14	6.00	6.00	632	642	10.00	10.00	0.000	0.000	0.974	1.21
15	6.00	6.00	634	646	10.00	10.00	0.000	0.000	1.01	1.31
16	6.00	6.00	630	644	10.00	10.00	0.000	0.000	0.933	1.29
17	6.00	6.00	625	632	10.00	10.00	0.000	0.000	0.784	0.989
18	6.00	6.00	618	622	10.00	10.00	0.000	0.000	0.573	0.715
19	6.00	6.00	617	622	10.00	10.00	0.000	0.000	0.573	0.695

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.